

EZHILAN CHINNASAMY

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SUMMARY

Full-stack data professional with fintech and consulting expertise, integrating analytics, BI storytelling, and data engineering to generate measurable impact. Proficient in SQL, Python, and Azure Data Factory, with experience creating automated ETL pipelines, executive-ready dashboards, and predictive models that drive operational efficiency and revenue growth. AI enthusiast, actively expanding skills through applied projects in agentic AI, autonomous workflows, and AI-driven decision systems. Recognized for rapid execution and translating data complexity into actionable strategy.

SKILLS

Core Data & Programming: SQL (MySQL, PostgreSQL, SQL Server), Python, R, Excel VBA, Git, Linux

Data Engineering & Cloud: Azure Data Factory (ADF), ETL/ELT Pipelines, Data Warehousing, Hadoop, PySpark, Apache Kafka, Databricks

BI & Visualization: Looker Studio, Power BI, Tableau, Advanced Excel (Pivot Tables, LOOKUP, Power Query), Google Analytics, Adobe Analytics

AI / Machine Learning: Predictive Modeling, Regression & Classification, Clustering, Time Series Forecasting, NLP (TF-IDF, VADER, BERT), Ensemble Models, Agentic AI, Autonomous Workflows, AI-Driven Decision Systems, Prompt Engineering for AI Agents

Data Strategy & Collaboration: Data Storytelling, Stakeholder Management, Process Optimization, Cross-Functional Collaboration, Zendesk

WORK EXPERIENCE

Pay with Spire, Data Analyst (Looker Studio, MS SQL, Azure Data Factory, Excel, Python)

Sep 2024 – present | Dallas, TX

- Operated across the data lifecycle blending analyst precision, BI storytelling, and engineering rigor during a major product expansion, delivering insights that shaped executive strategy and brought operational clarity across partner and internal teams.
- Deployed 15+ Looker Studio dashboards to track KPIs, partner performance, and trends, becoming a core tool in weekly leadership reviews.
- Engineered fully automated ETL pipelines pulling scheduled exports from Zendesk, processing with Python scripts in Azure Runbook, and loading into BUSSQL via Azure Data Factory, reducing manual workload by 40% and cutting reporting turnaround from 2 days to under 4 hours.
- Embedded predictive models and advanced analytics into partner reports, enabling proactive risk detection and targeted marketing campaigns that improved conversion rates by 12% while streamlining cross-team workflows to shorten project delivery timelines by 15%.

The University of Texas at Dallas, Student Assistant (Figma, React, Python, SQL, AI)

Jun 2024 – Sep 2024

- Oversaw web development of a data-driven interactive timeline, guiding a 3-member team in sourcing, structuring, and presenting historical data.
- Directed data analysis using Python and SQL to mine decades of Texas tech records, integrating exploratory analytics and AI-assisted workflows that uncovered innovation patterns and produced a scalable framework for future university visualization projects.
- Delivered a responsive timeline in Figma and React that improved accessibility to 50+ years of historical datasets and established a scalable framework for future university visualization projects.

Ernst & Young Global Delivery Services, Software Engineer

Feb 2021 – Oct 2022

(Adobe Analytics, Python, JavaScript, Git, Azure, React.js, Java)

- Directed the execution of Adobe Analytics across 50+ websites using Tag Management System, leveraging advanced features like segments, metrics, and funnels to create insightful reports and interactive dashboards, resulting in a significant pipeline value increase from 65.4% to 89.32%.
- Developed automated Python workflows for the Service Desk, improving data accuracy by 50% and cutting processing time by 80%, while enabling faster resource allocation and support analytics.
- Designed user interfaces for a food ordering application using React, HTML, and CSS, leading to a 40% increase in user engagement.

PROJECTS

Curse of Dimensionality - Mercedes-Benz Greener Manufacturing,

May 2024 – Jul 2024

(Feature Engineering, Ensemble - Stacking, Model building)

- Developed an ML algorithm using NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn achieving a 35% improvement in the prediction.
- Performed comprehensive feature engineering, hyperparameter tuning, grid search, and cross-validation to optimize model performance.
- Employed machine learning algorithms like Random Forest, XGBoost, Lasso Regression, SVM, and Gradient Boosting, finally trained a Meta learner (stacking - RF) achieving an R-squared (R^2) score of 0.7, MSE of 0.14 and MAE of 0.24 across all models.

Cold Start E-Commerce Product Recommendation System,

Feb 2024 – Apr 2024

(Unsupervised Learning, NLP, Clustering)

- Constructed a model-based collaborative filtering system using Truncated SVD and cosine similarity to recommend products based on user purchase history, achieving a recommendation coverage of 94.92% and an RMSE of 0.8256.
- Designed a content-based recommendation system leveraging TF-IDF vectorization and K-Means clustering to suggest products based on textual analysis of product descriptions, optimizing clustering with silhouette scores.
- Devised Cold Start recommender using VADER and BERT models on new users' microblogging to infer their preferences.

EDUCATION

The University of Texas at Dallas, MS in Business Analytics and Artificial Intelligence

Jan 2023 – Dec 2024 | GPA: 3.7

Coursework: Business Analytics with R, Advanced Statistics, Database Foundations, Prescriptive Analytics, Data Visualization, Applied Machine Learning, Big Data, Programming for Data Science, Predictive Analytics, Applied Deep Learning, Applied Econometrics and Time Series Analysis.

Certifications issued by UTD: Applied Machine Learning Certificate, Business Decision Analytics Certificate.

Kumaraguru College of Technology, Bachelors in Computer Science and Engineering

Jul 2017 – Jun 2021 | GPA: 3.7

PUBLICATION

Ezhilan. Smart Parking , International Conference on Advancements in Electrical, Electronics, Communication, Computing and Automation(2021).